WILLIAM MUELLER

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Education

UNIVERSITY OF VIRGINIA – Charlottesville, VA **Expected Graduation: May 2024**

B.A. Computer Science:

Cumulative GPA: 3.87/4.00

Relevant Coursework:

Algorithms, Program & Data Representation, Discrete Mathematics, Software Dev Methods, Cybersecurity, Computer Architecture

Work Experience

RESEARCH INTERN – *UVA Biocomplexity Institute*, Charlottesville, VA **June 2022 - Present**

- Developed a model to represent the spread of the tomato pest *Tuta Absoluta* in the United States
- Worked under the mentorship of Dr. Abhijin Adiga, an expert in network science
- Implemented 5 different dispersal kernel models to improve Dr. Adiga's previous simulator
- Performed regressions, correlations, and other statistical tests using python libraries like pandas and scikit-learn
- Generated spatial and temporal disaggregations of tomato flows based on various datasets
- Presented various findings to researchers and other interns in a symposium hosted by the Biocomplexity Institute

DECKHAND – *Potomac Riverboat Company*, Alexandria, VA **May 2021 - Aug 2021**

- Secured boats to docks with lines, cast off lines to enable departure, and assisted the captain and other deckhands
- Welcomed 100+ guests to boats daily, answered questions, and served food and beverages

Relevant Projects

A* SEARCH ALGORITHM VISUALIZATION

- Researched the A* search algorithm and replicated it in Java with a GUI while utilizing Java's SwingWorker class

CHESS VISUALIZATION TOOL

- Designed and hosted a website which can be deployed as a tool to improve a chess player's visual memory
- Integrated JavaScript libraries such as chessboardjs and Bootstrap, and created a Firebase backend database

STUDY BUDDY FINDER

- Served as DevOps Manager on a team that created a fully functional Django app to help students find study buddies
- Performed biweekly sprints to model an Agile workflow with the team of 4 other students to gradually add features

Extracurriculars

PRES. OF TECHNOLOGY, TEAM MEMBER – HooHacks, University of Virginia's Annual Hackathon **Sept 2020 - Present**

- Controlling, updating, and managing multiple HooHacks websites, including the registration system
- Supervised $60,000 budget, created run of show, and managed prizes for Spring 2021 hackathon
- Handling the newsletter and authored newsletter article about embracing limitations in computer science projects
- Managed event with over 500 participants while working with team of 20 members

CHAIR – Bishop Ireton's Honor Council **Aug 2019 – May 2020**

- Oversaw hearings for reported honor code violations while working with the administration

DIRECTOR, ASSISTANT DIRECTOR, TEAM MEMBER – HackBI, Bishop Ireton's Annual Hackathon **Nov 2017 – Feb 2020**

- Led team of 25 members and over 30 adult volunteers throughout 24-hour event that had 150 participants
- Supervised $10,000 budget and created run of show
- Contacted and corresponded with school administration and event sponsors
- Judged final projects, approved prizes, filled out purchase orders, and organized and led weekly team meetings

Skills

Programming:	**Web:**	**Other:**
Java, Python, C++	Django, HTML, CSS, Javascript, React	Git, Github, Blender, Firebase, SQL